P1 Wave packet

$$\vec{E}(x,0) = \vec{E}_0\, e^{i k_0 x}\, e^{-\frac{x^2}{2\ell^2}}$$

$$\vec{E}(\vec{k},0) = \int d^{\cancel{3}}\cancel{x}\; \vec{E}(x,0)\, e^{-ikx} = \vec{E}_0 \int_{-\infty}^{\infty} dx\; e^{i(k_0-k)x}\, e^{-\frac{x^2}{2\ell^2}}$$

$$= (x=\ell y) = \vec{E}_0\, \ell \int dy\; e^{i(k_0-k)\ell y}\, e^{-\frac{1}{2}y^2}$$

$$-\frac{1}{2}y^2 + i(k_0-k)\ell y = -\frac{1}{2}\left\{\left(y - i(k_0-k)\ell\right)^2 + (k_0-k)^2\ell^2\right\}$$

$$\vec{E}(k,0) = \vec{E}_0\, \ell \underbrace{\int_{-\infty}^{\infty} dy\; e^{-\frac{1}{2}(y+i(k-k_0)\ell)^2}}_{\sqrt{2\pi}}\; e^{-\frac{1}{2}(k-k_0)^2\ell^2}$$

$$= \sqrt{2\pi}\,\ell\, \vec{E}_0\, e^{-\frac{1}{2}(k-k_0)^2\ell^2}$$

Gaussian distribution in $\vec{k}$-space!

$$\vec{E}(k,t) = \sqrt{2\pi}\,\ell\, \vec{E}_0\, e^{-\frac{1}{2}(k-k_0)^2\ell^2}\, e^{-i\omega(k)t}$$

$$\vec{E}(x,t) = \int_{-\infty}^{\infty} \frac{dk}{2\pi}\; \vec{E}(k,t)\, e^{ikx} = \sqrt{2\pi}\,\ell\, \vec{E}_0 \int \frac{dk}{2\pi}\; e^{-\frac{1}{2}(k-k_0)^2\ell^2} \cdot$$

$$\cdot\; e^{-i\omega(k)t} \cdot e^{i(k-k_0)x} \cdot e^{ik_0x}$$

$$(k=k_0+p) \quad = \sqrt{2\pi}\,\ell\, \vec{E}_0\, e^{ik_0x} \int_{-\infty}^{\infty} \frac{dp}{2\pi}\; e^{-\frac{1}{2}p^2\ell^2}\, e^{+ipx}\, e^{-i\omega(k_0+p)t}$$

let us expand $\omega(p+k_0) \approx \omega(k_0) + \omega'(k_0)\,p + \frac{1}{2}\omega''(k_0)\,p^2$

$$= \omega_0 + \omega_0'\,p + \frac{1}{2}\omega_0''\,p^2$$

$$\Rightarrow \vec{E}(x,t) = \sqrt{2\pi}\,\ell\, \vec{E}_0\, e^{ik_0x}\, e^{-i\omega_0 t} \int_{-\infty}^{\infty} \frac{dp}{2\pi}\; e^{ipx - \frac{1}{2}p^2\ell^2} \cdot$$

$$\cdot\; e^{-i\left(\omega_0' p + \frac{1}{2}\omega_0'' p^2\right)t}$$

We need to evaluate

$$J = \int_{-\infty}^{\infty} \frac{dp}{2\pi} \; e^{ip\underbrace{(x - \omega_0' t)}_{\beta}} \; e^{-\frac{1}{2}p^2 \underbrace{(\ell^2 + i\omega_0'' t)}_{\alpha}}$$

$$-\frac{1}{2}\alpha p^2 + i\beta p = -\frac{1}{2}\alpha\left(p^2 - 2\frac{i\beta}{\alpha}p\right) =$$

$$= -\frac{1}{2}\alpha\left\{\left(p - \frac{i\beta}{\alpha}\right)^2 + \frac{\beta^2}{\alpha^2}\right\} = -\frac{1}{2}\alpha\left(p - \frac{i\beta}{\alpha}\right)^2 - \frac{\beta^2}{2\alpha}$$

$$\Rightarrow \; J = e^{-\frac{\beta^2}{2\alpha}} \int_{-\infty}^{\infty} \frac{dp}{2\pi} \; e^{-\frac{\alpha}{2}\left(p - \frac{i\beta}{\alpha}\right)^2} =$$

$$= e^{-\frac{\beta^2}{2\alpha}} \frac{\sqrt{2\pi}}{\sqrt{\alpha}} \frac{1}{2\pi}$$

$$\vec{E}(x,t) = \frac{\ell \vec{E}_0 \, e^{i(k_0 x - \omega_0 t)}}{\sqrt{\ell^2 + i\omega_0'' t}} \exp\left\{-\frac{1}{2}\cdot\frac{(x - \omega_0' t)^2}{(\ell^2 + i\omega_0'' t)}\right\}$$

$$= \vec{E}_0 \, e^{i(k_0 x - \omega_0 t)} \frac{\ell}{\sqrt{\ell^2 + i\omega_0'' t}} \exp\left\{-\frac{(x - \omega_0' t)^2 (\ell^2 - i\omega_0'' t)}{\ell^4 + (\omega_0'' t)^2}\right\}$$

The packet moves with $v_g = \omega_0' = \left.\frac{d\omega}{dk}\right|_{k_0}$ = group velocity.
Its width increases as

$$\ell(t) = \sqrt{\ell^2 + \left(\frac{\omega_0'' t}{\ell}\right)^2}$$

therefore it becomes broader with rate $\frac{\omega_0'' t}{\ell}$.

P2 $\vec{\nabla}\cdot\vec{D} = \rho_{ext}(\vec{r},t)$

$$\vec{D}(\vec{r},t) = \int \frac{d^3\vec{k}}{(2\pi)^3}\int\frac{d\omega}{2\pi}\,\vec{D}(\vec{k},\omega)\,e^{i\vec{k}\cdot\vec{r}-i\omega t}$$

$$\rho_{ext}(\vec{r},t) = \int\frac{d^3\vec{k}}{(2\pi)^3}\int\frac{d\omega}{2\pi}\,\rho_{ext}(\vec{k},\omega)\,e^{i\vec{k}\cdot\vec{r}-i\omega t}$$

$$\Rightarrow\quad i\vec{k}\cdot\vec{D}(\vec{k},\omega) = \rho_{ext}(\vec{k},\omega)$$

But $\vec{k}\cdot\vec{D} = i k\varepsilon E$ where $\varepsilon(k,\omega)$ = longitudinal dielectric function

$$E(\vec{k},\omega) = \frac{\rho_{ext}(\vec{k},\omega)}{i k\,\varepsilon(k,\omega)}$$

$$\vec{E} = -\vec{\nabla}\varphi \Rightarrow \vec{E}(k,\omega) = -i\vec{k}\,\varphi(\vec{k},\omega)$$

$$\Rightarrow \varphi(\vec{k},\omega) = \frac{E(\vec{k},\omega)}{-ik} = \frac{\rho_{ext}(\vec{k},\omega)}{k^2\varepsilon(k,\omega)}$$

In our problem $\rho_{ext}(\vec{r},t) = Q\,\delta(\vec{r})$ (t-independent!)

$$\Rightarrow \rho(\vec{k},\omega) = \int d^3\vec{r}\int dt\; e^{i\omega t - i\vec{k}\cdot\vec{r}}\,Q\,\delta(\vec{r}) =$$

$$= Q\int dt\, e^{i\omega t} = 2\pi Q\,\delta(\omega)$$

Thus we obtain that

$$\varphi(\vec{k},\omega) = \frac{1}{k^2\varepsilon(k,\omega)}\,2\pi Q\,\delta(\omega) = \frac{2\pi Q\,\delta(\omega)}{k^2\varepsilon(k,0)}$$

We need $\varepsilon_\ell(k,0) = \varepsilon_{zz}(k,0)$ at zero frequency.
In lecture 17 we found that for $\omega \ll v_T k$

$$\varepsilon_{zz}(k,\omega) = \varepsilon_0 + \varepsilon_0\frac{\omega_p^2}{(v_T k)^2}\left(1 - \left(\frac{\omega}{v_T k}\right)^2 + i\sqrt{\frac{\pi}{2}}\,\frac{\omega}{v_T k}\right)$$

$$v_T \equiv \sqrt{\frac{k_B T}{m}},\quad \omega_p^2 = \frac{ne^2}{\varepsilon_0 m}$$

Therefore $\varphi(\vec{k},\omega) = \dfrac{2\pi Q\,\delta(\omega)}{\varepsilon_0}\,\dfrac{1}{k^2\left(1+\dfrac{\omega_p^2}{v_T^2 k^2}\right)}$

$$\varphi(\vec{r},t) = \int \frac{d^3\vec{k}}{(2\pi)^3}\int\frac{d\omega}{2\pi}\,\varphi(\vec{k},\omega)\,e^{i\vec{k}\cdot\vec{r}-i\omega t} =$$

$$= \int \frac{d^3\vec{k}}{(2\pi)^3}\,\frac{Q}{\varepsilon_0}\,\frac{e^{i\vec{k}\cdot\vec{r}}}{k^2+\dfrac{\omega_p^2}{v_T^2}} =$$

$$= \frac{Q}{(2\pi)^2\varepsilon_0}\int_0^\infty dk\,k^2\underbrace{\int_0^{2\pi} d\varphi}_{2\pi}\underbrace{\int_0^{\pi} d\theta\,\sin\theta}_{-\int_1^{-1} d(\cos\theta)}\,\frac{e^{ikr\cos\theta}}{k^2+\dfrac{\omega_p^2}{v_T^2}}$$

$$= \frac{Q}{(2\pi)^2\varepsilon_0}\int_0^\infty dk\,\frac{k^2}{k^2+\dfrac{\omega_p^2}{v_T^2}}\underbrace{\int_{-1}^{1} du\,e^{ikru}}_{\frac{e^{ikru}}{ikr}\Big|_{-1}^{1} = \frac{e^{ikr}-e^{-ikr}}{ikr} = \frac{2i\sin(kr)}{ikr}}$$

$$= \frac{Q}{(2\pi)^2\varepsilon_0}\,\frac{2}{r}\int_0^\infty dk\,\frac{k\sin(kr)}{k^2+\dfrac{\omega_p^2}{v_T^2}}$$

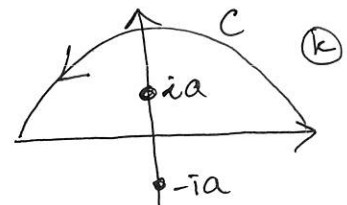
The integrand is even function of k $\Rightarrow$ the integral can be written as

$$\frac{1}{2}\int_{-\infty}^{\infty} dk\,\frac{k\sin(kr)}{k^2+a^2} \qquad \left(a^2 = \frac{\omega_p^2}{v_T^2}\right)$$

Note that here $r = |\vec{r}| > 0$

Thus $\varphi(\vec{r},t) = \frac{Q}{(2\pi)^2 \varepsilon_0 r} \left(-\frac{d}{dr}\right) \int_{-\infty}^{\infty} dk \, \frac{\cos(kr)}{k^2+a^2}$

$$I = \mathrm{Re} \int_{-\infty}^{\infty} dk \, \frac{e^{ikr}}{k^2+a^2} =$$



$$= \mathrm{Re} \oint_C dk \, \frac{e^{ikr}}{k^2+a^2} = \mathrm{Re} \, \frac{2\pi i}{2ia} e^{ir(ia)} = \frac{\pi}{a} e^{-ra}$$

$$\varphi(\vec{r},t) = \frac{Q}{4\pi^2 \varepsilon_0 r} \left(-\frac{d}{dr}\right) \frac{\pi}{a} e^{-ra} = \frac{Q}{4\pi^2 \varepsilon_0 r} \pi e^{-ra}$$

$$= \frac{Q}{4\pi \varepsilon_0 r} e^{-r/\ell_D}$$

where $\ell_D = \frac{1}{a} = \frac{v_T}{\omega_p} = \sqrt{\frac{k_B T}{m} \cdot \frac{\varepsilon_0 m}{n e^2}} = \sqrt{\frac{\varepsilon_0 k_B T}{e^2 n}}$

is Debye radius = distance on which Coulomb potential Q is exponentially screened.

Note that $\varphi(\vec{r},t)$ is in fact t-independent, as it of course should be for the static charge.

Problem 3. The equation for the drift velocities are

$$\frac{d\vec{v}_1}{dt} = -\frac{e}{m_1}\vec{E} - \frac{e}{m_1}\vec{v}_1 \times \vec{B} - \frac{\vec{v}_1}{\tau} = 0,$$

$$\frac{d\vec{v}_2}{dt} = \frac{e}{m_2}\vec{E} + \frac{e}{m_2}\vec{v}_2 \times \vec{B} - \frac{\vec{v}_2}{\tau} = 0,$$

each equation is solved independently, similar to how it was done in class, then $\vec{j} = -en_1\vec{v}_1 + en_2\vec{v}_2$ The total conductivity is the sum of the two conductivities

$$\sigma_{ik} = \sigma_{ik}^{(1)} + \sigma_{ik}^{(2)}$$

where $$\sigma_{ik}^{(1)} = \frac{e^2 n_1 \tau_1}{(\omega_{c1}^2\tau_1^2 + 1)m_1}\begin{pmatrix} 1 & -\omega_{c1}\tau \\ \omega_{c1}\tau & 1 \end{pmatrix}$$

where $\omega_{c1} = \frac{eB}{m_1} > 0$

Similarly,

$$\sigma_{ik}^{(2)} = \frac{e^2 n_2 \tau_2}{(\omega_{c2}^2 \tau^2 + 1) m_2} \begin{pmatrix} 1 & \omega_{c2}\tau \\ -\omega_{c2}\tau & 1 \end{pmatrix} \qquad \omega_{c2} = \frac{eB}{m_2} > 0$$

The Hall current (perpendicular to both $\vec{E}$ and $\vec{B}$) is given by off-diagonal terms

$$j_H = \frac{-e^2 n_1 \tau_1^2 E \omega_{c1}}{(\omega_{c1}^2 \tau^2 + 1) m_1} + \frac{e^2 n_2 \tau_2^2 \omega_{c2} E}{(\omega_{c2}^2 \tau^2 + 1) m_2}$$

For strong magnetic field, such that

$$\omega_{c1}\tau \gg 1 \quad \text{and} \quad \omega_{c2}\tau \gg 1$$

the Hall current simplifies to

$$j_H = -\frac{e^2 n_1}{m_1 \omega_{c1}} E + \frac{e^2 n_2}{m_2 \omega_{c2}} = \frac{e(n_2 - n_1)E}{B}$$

We see that for equal number of electrons and holes (compensated semiconductor) the Hall effect vanishes. Note that for n-type and p-type semiconductors the Hall current has <u>opposite</u> sign.